# No connected graph with at most ten vertices is an unweighted death-Birth amplifier of weak selection: an exhaustive census, and a Lean check that Richter's eleven-vertex graph is one

Xavier Callens (SocrateAI Lab)

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**AI-assistance disclosure.** This preprint, its Lean code, its Python code and its claim ledger were produced by AI agents (Claude, Anthropic) under the author's direction. The author is responsible for the content. No human has audited the Lean statements for adequacy, and no human referee has checked the hand arguments. An **independent AI audit agent** (producer ≠ verifier, 2026-10-27; `AUDIT.md`) returned the verdict **ACCEPT_WITH_CORRECTIONS**, with no blocking item; its corrections are applied in this version (`REVISION.md`). It recompiled all Lean modules into a private directory (oleans byte-identical to the producer's) and replayed the kernel with `leanchecker` on `Core`, `DataR11`, `DataP11`, `Richter11` and `Controls` (rc 0 each); it rechecked the census with its own exact method (all graphs for $N = 3..9$ and a 20 177-graph sample at $N = 10$) and reproduced the finite-$r$ signs. **No human audit exists, and no statement lock exists.**

**Abstract**

For death-Birth (dB) updating on a connected, simple, unweighted graph $G$ with $N$ vertices, Allen et al. showed that $\rho_G(1+\delta) = \frac{1}{N} + \delta\,\frac{N_{\mathrm{eff}}-2}{2N} + O(\delta^2)$, with $N_{\mathrm{eff}}$ an effective population size computed from coalescence times of random walks, and called $G$ an *amplifier of weak selection* if $N_{\mathrm{eff}} > N$. Richter (2020) exhibited an eleven-vertex example. We enumerated *every* connected graph with $3 \le N \le 10$ (2, 6, 21, 112, 853, 11 117, 261 080 and 11 716 571 graphs) and found **none** with $N_{\mathrm{eff}} > N$: every regular graph has $N_{\mathrm{eff}} = N$ exactly and every other graph has $N_{\mathrm{eff}} < N$, the closest non-regular graph at $N = 10$ lying at least 0.0207 below $N$. In Lean 4 we check that on Richter's graph *every* rational solution of Allen's coalescence system has $N_{\mathrm{eff}} = 2668552698098/242579025123 \approx 11.000756 > 11$. Together, *within the weak-selection criterion for unweighted graphs*, the smallest dB amplifier has exactly 11 vertices.

**Tier split.** [A] (Lean 4.34.1, axioms within {`propext`, `Classical.choice`, `Quot.sound`}): the theorem `R11_amplifier` (above), the uniqueness theorem `eq5_unique` (a positive solution of the system is the only solution), the isothermal control `P11_isothermal` ($N_{\mathrm{eff}} = 11$ on the quartic parent), and kernel-decided negative controls. [L] (literature, not re-proved): the criterion $N_{\mathrm{eff}} > N$, Allen's coalescence equations, and the step from $N_{\mathrm{eff}} > N$ to $\rho_G > \rho_{K_N}$ for $r$ slightly above 1 (their eq. (8)). [C]: the census itself (enumeration with nauty `geng`, counts equal to OEIS A001349; floating-point proposals decided by an exact integer residual bound; exact `Fraction` solves for the regular graphs), and exact-rational sign certificates for $\rho_G - \rho_{K_{11}}$ at four values of $r$. The census is *not* kernel-checked. [X]: finite-$r$ numerics beyond the certified points (extrapolated windows, floating-point ratios). Every claim in the text carries one of these tags and a ledger identifier (`AMC-·-nnnn`).

**What this is NOT.** This is mathematics of graphs and of fixation probabilities in a Moran-type model, with the single-mutant, weak-selection criterion of Allen et al. It is **not** a statement about real tumours, tissues, patients or any medical application: no biological structure was studied, no data were used, and nothing here bears on diagnosis, treatment, dose or schedule. It is also not a statement about weighted or directed graphs, other update rules, or selection of finite strength.

**Markers and ledger.** [A]: kernel-checked Lean, axioms within the allowlist, controls caught. [L]: a cited source, taken as stated. [C]: executed code with an argument that is written down but not machine-checked. [X]: floating point, supporting nothing by itself. A [A] tag attaches to the Lean statement, never to its biological or physical gloss. Identifiers `AMC-··-nnnn` refer to `ledger/ledger.json` (Section 7).

# 1 Setting, predecessor and preregistration

This is item C1 of a computational programme (Proposal 4, preregistered 2026-10-27 in `results/proposal4/2026-10-27/preregistration.json`). It continues an earlier kernel-checked paper on the *isothermal theorem* for birth-death updating [11]: there, a population on an isothermal graph (equal column and row sums of the replacement matrix) fixes mutants exactly as the well-mixed population does, for every fitness $r$. For death-Birth updating the analogous statement is only first order: regular graphs tie with the complete graph at order $\delta$, and the question becomes which graphs do *better* (amplifiers) or *worse* (suppressors) than a well-mixed population when selection is weak. Allen et al. [1] gave an exact first-order criterion; Richter [2, 3] and Svoboda et al. [7] describe the eleven-vertex example. Svoboda et al. (p. 5) call it "the smallest known unweighted dB-amplifier"; Richter (2020, p. 6) writes that "the smallest graph for which $N_{\text{eff}} > N$ was found is the quartic graph with size $N = 11$", meaning within his regular-input search, and Richter (2022, p. 8) reports that no transient amplifier was found for regular graphs with $N \leq 10$ using his method. None of these is an exhaustive statement over all graphs. The preregistered item C1b asked: *either no connected unweighted graph with $N \leq 10$ has $N_{\text{eff}} > N$, or the exact list of those that do.* The kill rules were: if the $N = 10$ census exceeds two budgets, report $N \leq 9$; if the criterion's hypotheses cannot be pinned from the primary source, the census stays Tier X. Neither fired: the $N = 10$ enclosure took 1237 s of one core (plus the candidate solves), and the criterion was read in Allen et al. (arXiv v1, definitions on pp. 6 and 8; eqs. (5)–(9) and the definitions are identical in the published version, checked by the auditor). Prior 0.45 for a definite census; the result is a definite (negative) census.

# 2 A step-by-step account for non-specialists

**Step 1: the death-Birth process.** [L] A graph $G$ has $N$ vertices, each occupied by one individual: a *resident* (fitness 1) or a *mutant* (fitness $r = 1 + \delta$). In one step a vertex $i$ is chosen *uniformly at random to die*; each neighbour $j$ of $i$ then competes to fill the vacancy with probability proportional to its fitness, $f_j / \sum_{k \sim i} f_k$, and the winner places a copy of itself on $i$. Start with one mutant at a uniformly random vertex. The process ends when the mutant type has taken over (*fixation*) or died out. The *fixation probability* $\rho_G(r)$ is the chance of fixation. For the complete graph $K_N$ it is an explicit formula; when $r = 1$ every graph has $\rho = 1/N$. (Lean: the process is *not* formalised; only the linear system of Step 3 is.)

**Step 2: weak selection, and what "amplifier" means here.** [L] Write $r = 1 + \delta$. Allen et al. show

$$\rho_G(1 + \delta) = \frac{1}{N} + \delta \, \frac{N_{\text{eff}} - 2}{2N} + O(\delta^2) \qquad \text{(their eq. (8))}, \tag{1}$$

so for $K_N$ (where $N_{\text{eff}} = N$) the slope is $(N - 2)/2N$, which agrees with the exact formula $\frac{N-1}{N} \frac{1-1/r}{1-r^{1-N}}$ (a check we made). Hence $\rho_G(1 + \delta) - \rho_{K_N}(1 + \delta) = \delta \, (N_{\text{eff}} - N)/(2N) + O(\delta^2)$. If $N_{\text{eff}} > N$ then $G$ beats $K_N$ for all small $\delta > 0$ and loses for small $\delta < 0$: $G$ is an *amplifier of weak selection*. If $N_{\text{eff}} < N$ it is a *suppressor*. If $N_{\text{eff}} = N$ the first order decides nothing. Notice what this is not: the statement concerns $r$ close to 1 and an unknown threshold $b > 1$ beyond which the sign may flip; it does not say $G$ amplifies for every $r > 1$.

**Step 3: coalescence times and $N_{\text{eff}}$.** [L] Follow two lineages backward in time. Each step, one of the two (chosen with probability $\frac{1}{2}$) jumps to a uniformly random neighbour: from $i$ it goes to $k$ with probability $p_{ik} = 1/k_i$ for $k \sim i$, where $k_i$ is the degree. Let $\tau_{ij}$ be the expected time until they meet, with $\tau_{ii} = 0$. Then (Allen eq. (5), `AmplifierCensus.Eq5`)

$$\tau_{ij} = 1 + \tfrac{1}{2} \sum_k \big( p_{ik} \tau_{jk} + p_{jk} \tau_{ik} \big) \quad (i \neq j). \tag{2}$$

The *remeeting time* is $\tau_i = 1 + \sum_j p_{ij}\tau_{ij}$ (eq. (6), `remeet`), and with $\pi_i = k_i / \sum_j k_j$

$$N_{\text{eff}} = \sum_i \pi_i \tau_i \qquad \text{(eq. (9), } \texttt{neff}\text{)}. \tag{3}$$

*Worked example* (exact, `data/worked_example.py`). Path on three vertices $0-\{1,2\}$ (a star $S_2$, centre 0): by symmetry $\tau_{01} = \tau_{02}$ and (2) gives $\tau_{01} = 1 + \frac{1}{4}\tau_{12}$, $\tau_{12} = 1 + \tau_{01}$, hence $\tau_{01} = \frac{5}{3}$, $\tau_{12} = \frac{8}{3}$, all $\tau_i = \frac{8}{3}$ and $N_{\text{eff}} = \frac{8}{3} < 3$: a suppressor (it equals $4n/(n+1)$ at $n = 2$, Allen's star). The triangle $K_3$ gives all $\tau_{ij} = 2$, $\tau_i = 3$, $N_{\text{eff}} = 3 = N$. The graph $K_4$ minus an edge gives $N_{\text{eff}} = 173/45 \approx 3.844 < 4$. Two identities are used by the code (checked exactly on all 994 connected graphs with $3 \le N \le 7$, `AMC-C-0006`): $\sum_i \pi_i^2 \tau_i = 1$ (eq. (7)), and, because $\pi_i p_{ij} = 1/2m$ for $m$ edges,

$$N_{\text{eff}} = 1 + \frac{1}{m} \sum_{\{i,j\} \in E} \tau_{ij}. \tag{4}$$

**Step 4: why regular graphs tie.** [L] [C] If every degree equals $k$, then $\pi_i = 1/N$, and eq. (7) reads $\frac{1}{N^2}\sum_i \tau_i = 1$, so $N_{\text{eff}} = \frac{1}{N}\sum_i \tau_i = N$ exactly. This is the first-order isothermal theorem. The census confirms it for all $1+2+2+5+4+17+22+167$ connected regular graphs with $3 \le N \le 10$ and shows no non-regular graph attains equality (`AMC-C-0004`). In Lean, the same fact on the quartic parent of Richter's graph is `P11_isothermal` ([A], `AMC-A-0003`): $N_{\text{eff}} = 11$ exactly.

**Step 5: deciding $N_{\text{eff}} > N$ for 11.7 million graphs without trusting a float.** [C] Solving (2) exactly for every graph with $N = 10$ would cost hours per $10^5$ graphs. Instead, a floating-point solve only *proposes*; an exact integer computation *decides*.

1. Eliminate $\tau_{ii} = 0$ and multiply (2) by $2k_ik_j$. The unknown pair-vector $\tau$ then satisfies an integer system $A_s\tau = b$, equivalently $(I - M)\tau = \mathbf{1}$ with $M \ge 0$ entrywise (entries $\frac{1}{2}p$) and row sums at most 1 (mass that lands on a diagonal pair is lost: the lineages have met).
2. Because the lineages of a connected graph do meet, $M^n \to 0$ and $(I-M)^{-1} = \sum_n M^n \ge 0$ entrywise; and $(I-M)^{-1}\mathbf{1} = \tau$ is the expected meeting-time vector.
3. Let $\tilde\tau$ be any approximate solution (here the float solution rounded to a multiple of $2^{-30}$). Its *residual* $\rho = (I-M)\tilde\tau - \mathbf{1}$ is computed exactly in 64-bit integers, giving $\varepsilon = \|\rho\|_\infty \le E/2^{31}$. Then $\tau - \tilde\tau = -(I-M)^{-1}\rho$, and since $(I-M)^{-1} \ge 0$,

$$|\tau_{ij} - \tilde\tau_{ij}| \le (I-M)^{-1}|\rho| \le \varepsilon\,(I-M)^{-1}\mathbf{1} = \varepsilon\,\tau_{\max}\ , \qquad \tau_{\max} \le \frac{\tilde\tau_{\max}}{1-\varepsilon}.$$

   Non-negativity of the inverse is the whole point: it turns a residual into a *componentwise* error bound without computing any norm of the inverse.
4. By (4), $|N_{\text{eff}} - \hat N_{\text{eff}}| \le \varepsilon\tau_{\max}$. A graph is *excluded* (certified $N_{\text{eff}} < N$) only if $\hat N_{\text{eff}} + \varepsilon\tau_{\max} < N$, tested in exact integers.
5. Every graph that is not excluded is solved exactly in rational arithmetic. In the census these were precisely the regular graphs (22 at $N = 9$, 167 at $N = 10$).

*Precondition.* The residual is formed in 64-bit integers, so the argument needs $\tilde\tau$ finite and $|t| = |\mathrm{rint}(2^{30}\tilde\tau)|$ small enough that no int64 overflow occurs (the code now asserts `isfinite` and $\max|t| < 2^{50}$; for realistic data $\tau \lesssim 10^2$ gives $|t| \le 2^{37}$ and row sums of $|A_s t| \lesssim 2^{50}$). A poor but finite proposal only enlarges $E$ and blocks an exclusion; it cannot cause a wrong one. The assert was added after the census run (`REVISION.md`); the independent audit recheck covers the decisions. The largest $\varepsilon$ seen was $5.9 \times 10^{-8}$ (at $N = 10$), while the smallest certified margin $N - N_{\text{eff}}$ among non-regular graphs was at least 0.0173 ($N = 9$; exactly 13/748) and 0.0207 ($N = 10$; exactly 13/627): the error budget is about six orders of magnitude below the decision margin. This argument is a hand proof implemented in Python; it is [C], not kernel-checked.

**Step 6: the Lean statement beside the data.** [A] Lean cannot reasonably enumerate $10^7$ graphs, but it can check one graph exactly. Let `Eq5 A tau` be (2) over $\mathbb{Q}$ for the adjacency $A$ on `Fin n`. `eq5_unique` says: *if $v$ solves `Eq5` and $v_{ij} > 0$ for $i \neq j$, then every rational solution $\tau$ equals $v$* (a maximum principle: the largest ratio $\delta_{ij}/v_{ij}$ of the difference $\delta = \tau - v$ would otherwise satisfy a strict inequality). So a single rational vector, produced by untrusted code and re-checked by the kernel with `decide +kernel`, certifies $N_{\text{eff}}$ for *all* solutions of the system, not for a guessed one. `R11_amplifier` then states

$$\forall \tau,\ \text{Eq5}\ G_{R11}\ \tau \Rightarrow N_{\text{eff}}(G_{R11}, \tau) = \tfrac{2668552698098}{242579025123}\ \wedge\ 11 < N_{\text{eff}}(G_{R11}, \tau).$$

The kernel also checks that the adjacency is symmetric, loopless, and has degrees 3 at vertices $0, 9$ and 4 elsewhere. What Lean does *not* check: that this linear system is the right one for the death-Birth process, that eq. (1) holds, or that the census is complete.

**Step 7: from $N_{\text{eff}} > N$ to finite $r$.** [L] [C] By (1) the step from $N_{\text{eff}} > N$ to "$\rho_G > \rho_{K_N}$ for $r \in (1, b)$" is literature (`AMC-L-0002`). To check that this is not an artefact of our reading of the criterion, we solved the exact $2^N - 2$-state chain for every connected graph with $N \leq 6$ (141 graphs) and 22 graphs (every 40th) at $N = 7$ (163 graphs in all) and found slope $(N_{\text{eff}} - 2)/2N$ in every case (`AMC-C-0006`). For the single graph with $N = 11$ we also certified signs of $\rho_G - \rho_{K_{11}}$ at chosen $r$ with the same device as Step 5: the transient-state matrix $Q$ of the 2046-state chain has $(I - Q)^{-1} \geq 0$ with row sums equal to expected absorption times $T$, so an exact residual bound gives $|x - \tilde{x}| \leq \|\rho_x\|_\infty T_{\max}$ with $T_{\max} \leq 242.43$ (`AMC-C-0011`).

# 3 Results

## 3.1 The census

Table 1: Connected simple graphs with $3 \leq N \leq 10$ (nauty `geng -c`), classified by $N_{\text{eff}}$ against $N$ (`AMC-C-0001`, `-0002`, `-0004`). Counts equal OEIS A001349; "$N_{\text{eff}} = N$" counts equal OEIS A005177 and all these graphs are regular. The last column is a certified lower bound on $N - N_{\text{eff}}$ over non-regular graphs ([C]), rounded down; the exact minimum gaps (audit recheck) are 7/45 at $N = 4$ and 13/748 at $N = 9$.

| $N$ | graphs | $N_{\text{eff}} > N$ | $N_{\text{eff}} = N$ (regular) | $N_{\text{eff}} < N$ | max $\varepsilon$ | smallest gap $N - N_{\text{eff}}$ |
|---|---|---|---|---|---|---|
| 3 | 2 | 0 | 1 | 1 | $4.7\times10^{-10}$ | 0.3333 |
| 4 | 6 | 0 | 2 | 4 | $2.8\times10^{-9}$ | 0.1555 |
| 5 | 21 | 0 | 2 | 19 | $1.1\times10^{-8}$ | 0.0753 |
| 6 | 112 | 0 | 5 | 107 | $1.0\times10^{-8}$ | 0.0612 |
| 7 | 853 | 0 | 4 | 849 | $1.9\times10^{-8}$ | 0.0318 |
| 8 | 11 117 | 0 | 17 | 11 100 | $3.3\times10^{-8}$ | 0.0330 |
| 9 | 261 080 | 0 | 22 | 261 058 | $4.5\times10^{-8}$ | 0.0173 |
| 10 | 11 716 571 | 0 | 167 | 11 716 404 | $5.9\times10^{-8}$ | 0.0207 |

$N = 1, 2$ are trivial ($K_2$ is regular, so $N_{\text{eff}} = 2$). The independent pure-`Fraction` implementation (no identity (4), no enclosure) was run on all 12 111 connected graphs with $N \leq 8$ and agrees with the enclosure census on every count and every classification (`AMC-C-0003`); an all-`Fraction` run at $N = 9$ was started and stopped (projected about 3 h of one core), so $N = 9, 10$ rest on the enclosure. The gap column is called `gap_upper` in the data files; it is a lower bound on $N - N_{\text{eff}}$ (a misnomer in the code).

## 3.2 What the nearest misses look like

The nearest non-regular graph is always *nearly complete*: for even $N$ the complete graph minus one edge, for odd $N$ the complete graph minus a maximum matching (one vertex keeps degree $N - 1$). It is not a sparse or a bridged graph. The sparse structures (cliques joined through bottlenecks) that Richter [3] identifies as the ingredients of amplifiers do not yet clear the bar below $N = 11$. This is an observation for $N \leq 10$; there is no proof for general $N$, and the descriptions are read from the graph6 strings of the data files, not proved by a separate argument.

Table 2: Non-regular graph closest to $N$ (smallest certified gap), [C] observation for $N \le 10$ (`AMC-C-0005`). Exact values are rational; the exact fractions at $N = 9, 10$ are from the independent audit recheck (the census files hold floating-point readings).

| $N$ | graph6 | description | $N_{\text{eff}}$ | gap bound |
|---|---|---|---|---|
| 3 | `BW` | path $P_3$ | 8/3 | 0.3333 |
| 4 | `C^` | $K_4$ minus an edge | 173/45 | 0.1555 |
| 5 | `D]{` | $K_5$ minus a maximum matching | 719/146 | 0.0753 |
| 6 | `E^~w` | $K_6$ minus an edge | 291/49 | 0.0612 |
| 7 | `F]~vw` | $K_7$ minus a maximum matching | 5038/723 | 0.0318 |
| 8 | `G^~~~{` | $K_8$ minus an edge | 2653/333 | 0.0330 |
| 9 | `H]~v~z~` | $K_9$ minus a maximum matching | $6719/748 \approx 8.98262$ | 0.0173 |
| 10 | `I^~~~~~~w` | $K_{10}$ minus an edge | $6257/627 \approx 9.97927$ | 0.0207 |

### 3.3 Richter's eleven-vertex graph

Graph6 `J?ouPjI[@i?`: the unique quartic (4-regular) graph on 11 vertices (among 265) from which an *amplifier constructor* edge removal works, minus one edge at the argmax-remeeting vertex. It has 21 edges and degrees $3, 4, \ldots, 4, 3$ (the two degree-3 vertices are non-adjacent). All four edge removals at that vertex give isomorphic graphs.

Table 3: Richter's $N = 11$ graph and the cubic $N = 12$ example ([C] except the Lean row; `AMC-A-0002`, `AMC-C-0009`, `-0010`).

| quantity | value here | Richter prints |
|---|---|---|
| $N_{\text{eff}}$, $N = 11$ (Lean: for *every* solution of Eq5) [A] | $2668552698098/242579025123 \approx 11.000756$ | 11.0008 |
| $N_{\text{eff}}$ of the quartic parent [A] | 11 exactly | — |
| $N_{\text{eff}}$, $N = 12$ cubic example [C] | $\approx 12.026644$ | 12.0266 |
| $\tau_{\max}$, $N = 11$ parent [C] | $10066/569 \approx 17.69069$ | 17.6997 |
| $\tau_{\max}$, $N = 12$ parent [C] | $\approx 18.29410$ | 18.2911 |
| margin $N_{\text{eff}} - N$ at $N = 11$ | $7.56 \times 10^{-4}$ | — |

The $\tau_{\max}$ values differ from the printed ones in the third decimal. The $N_{\text{eff}}$ values agree to every printed digit. This is an **unexplained discrepancy**: it may be a typo or a convention difference; it has not been raised with the author (`AMC-C-0010`). The independent audit confirmed our parent value ($10066/569 = 17.690685$ at vertex 9; 18.294102 for the $N = 12$ parent) and ruled out one explanation: Richter did not print the *perturbed* graph's value, since $\max_i \tau_i$ is 20.158 on the $N = 11$ amplifier and 22.067 on the $N = 12$ amplifier, and these match the printed numbers no better. The graph identification itself is solid (normalised-Laplacian $\lambda_2 = 0.0567$, mean degree 3.8182 as in Richter 2022 Fig. 1(a); $A_4(11) = 1$ in Richter 2020 Table 1; the four removals are pairwise isomorphic). The cause of the discrepancy remains unexplained.

### 3.4 Finite $r$ on the eleven-vertex graph

[C] Using the Step 7 device on the 2046-state dB chain, we certified (exact-rational bounds, total error $\le 3.3 \times 10^{-14}$): So $G$ beats $K_{11}$ just above $r = 1$ and loses just below it, exactly the pattern predicted by

Table 4: Certified intervals for $\rho_G(r) - \rho_{K_{11}}(r)$ on Richter's graph (`AMC-C-0011`).

| $r$ | interval | sign |
|---|---|---|
| $10001/10000 = 1.0001$ | $[3.02059 \times 10^{-9},\ 3.02065 \times 10^{-9}]$ | + |
| $2501/2500 = 1.0004$ | $[7.07914 \times 10^{-9},\ 7.07919 \times 10^{-9}]$ | + |
| $2499/2500 = 0.9996$ | $[-2.03974 \times 10^{-8},\ -2.03973 \times 10^{-8}]$ | − |
| $1001/1000 = 1.001$ | $[-7.3995 \times 10^{-9},\ -7.3994 \times 10^{-9}]$ | − |

$N_{\text{eff}} > N$ ([L]), and loses again at $r = 1.001$. Bisection between $r = 1.0007$ (+) and $r = 1.00085$ (−) with

certified signs at every step (`AMC-C-0012`) gives a certified sign change in $r \in [1.00082333, 1.00082349]$ (decimals rounded outward; exact $[10248431/10240000,\ 4099373/4096000]$): We certify signs at these points only: not

Table 5: The bisection (all signs certified, [C]).

| $r$ | decimal | sign | interval for $\rho_G - \rho_K$ |
|---|---|---|---|
| 10007/10000 | 1.0007 | + | $[3.6138 \times 10^{-9}, 3.6139 \times 10^{-9}]$ |
| 20017/20000 | 1.00085 | − | $[-9.4779 \times 10^{-10}, -9.4774 \times 10^{-10}]$ |
| 40031/40000 | 1.000775 | + | $[1.5691 \times 10^{-9}, 1.5692 \times 10^{-9}]$ |
| 16013/16000 | 1.0008125 | + | $[3.6969 \times 10^{-10}, 3.6974 \times 10^{-10}]$ |
| 160133/160000 | 1.00083125 | − | $[-2.7430 \times 10^{-10}, -2.7423 \times 10^{-10}]$ |
| 320263/320000 | 1.00082188 | + | $[5.1386 \times 10^{-11}, 5.1447 \times 10^{-11}]$ |
| 640529/640000 | 1.00082656 | − | $[-1.1053 \times 10^{-10}, -1.1047 \times 10^{-10}]$ |
| 256211/256000 | 1.00082422 | − | $[-2.9333 \times 10^{-11}, -2.9290 \times 10^{-11}]$ |
| 2562107/2560000 | 1.00082305 | + | $[1.1083 \times 10^{-11}, 1.1138 \times 10^{-11}]$ |
| 5124217/5120000 | 1.00082363 | − | $[-9.1128 \times 10^{-12}, -9.0596 \times 10^{-12}]$ |
| 10248431/10240000 | 1.0008233398 | + | $[9.829 \times 10^{-13}, 1.0493 \times 10^{-12}]$ |
| 4099373/4096000 | 1.00082349 | − | $[-4.0563 \times 10^{-12}, -4.0124 \times 10^{-12}]$ |

that there is exactly one crossing, and not the sign between them. **Second discrepancy with Richter.** Richter states $r_{\max} \approx 1.00075$ (text p. 8 and the Fig. 1 caption: "for the interval $1 < r < r_{\max} \approx 1.00075$ the fixation probability of the graph is larger ..."); this is the same crossing as our bracket. Our certified (+) signs at $r = 1.0007$, 1.000775 and 1.0008125 and the bracket [1.00082333, 1.00082349] place the first sign change above that value. The auditor independently certified (+) at $r = 1251/1250 = 1.0008$ ($\rho_G - \rho_K = 7.826 \times 10^{-10}$, error bound $2.5 \times 10^{-28}$) and at 16013/16000. Richter's value is approximate (read near the resolution of his Fig. 1b) and is inconsistent with the certified signs; the cause was not investigated (cf. $\tau_{\max}$ above). Beyond these points the "window $(1, 1.0008)$" is an [X] extrapolation (`AMC-X-0001`).

### 3.5 Regular graphs at finite $r$

[C] For all 212 connected non-complete regular graphs with $4 \le N \le 10$ we certified $\rho_G(r) < \rho_{K_N}(r)$ at both $r = 999/1000$ and $r = 1001/1000$ (none above, none undecided; `AMC-C-0013`). This is consistent with these ties being second-order *reducers* of fixation near $r = 1$, but it is **not** a proof for all $r$ close to 1; Allen et al. (p. 21) list it as open.

## 4 The headline, and its tiers

| statement | tier | basis |
|---|---|---|
| $N_{\text{eff}} > N$ iff amplifier of weak selection; first-order expansion (1) | [L] | Allen et al., arXiv v1, pp. 5–9, 32–34 (`AMC-L-0001`, `-0002`) |
| Any rational solution of Eq5 on $G_{R11}$ has $N_{\text{eff}} = 2668552698098/242579025123 > 11$ | [A] | `R11_amplifier` (`AMC-A-0002`) |
| A positive solution of Eq5 is the only solution | [A] | `eq5_unique` (`AMC-A-0001`) |
| Quartic parent has $N_{\text{eff}} = 11$ | [A] | `P11_isothermal` |
| No connected graph with $3 \le N \le 10$ has $N_{\text{eff}} > N$; regular $\Leftrightarrow N_{\text{eff}} = N$ | [C] | geng + OEIS counts + integer enclosure (`AMC-C-0001..0004`) |
| **The smallest connected unweighted dB amplifier of weak selection has exactly 11 vertices** | [C] | census (C) + Lean existence (A) + criterion (L) (`AMC-C-0014`) |
| Sign pattern of $\rho_G - \rho_{K_{11}}$ at four $r$, and a sign change in $[1.00082333, 1.00082349]$ | [C] | exact-rational residual bounds (`AMC-C-0011`, `-0012`) |
| Extrapolation of windows; floating ratios | [X] | `AMC-X-0001` |

The headline is filed Tier C because its census half is. The ledger gate would flag any claim filed above the tier of something it rests on.

# 5 Controls

Every control was run before any census count was reported (`data/controls.json`, `data/logs/controls.log`; `AMC-C-0006..0009`, `AMC-A-0003/0004`).

**Positive controls (all met).**

- $K_N$ and $C_N$, $N = 3..10$: $N_{\text{eff}} = N$ exactly (Allen's weak-selection isothermal theorem).
- All regular graphs have $N_{\text{eff}} = N$ exactly; their counts $1, 2, 2, 5, 4, 17, 22, 167$ equal OEIS A005177, so none was missing.
- Star $S_n$, $n = 2..9$: $N_{\text{eff}} = 4n/(n+1)$ exactly.
- Slope control: $d\rho/dr|_{r=1}$ of the exact $2^N$-state chain equals $(N_{\text{eff}} - 2)/(2N)$ on all 141 connected graphs with $N = 3..6$ and on 22 graphs (every 40th) at $N = 7$, 163 in total.
- Closed forms against the exact chain at $r \in \{\frac{3}{4}, \frac{5}{4}, \frac{3}{2}, \frac{7}{4}, \frac{1001}{1000}\}$, $N = 3..7$: dB and Bd on $K_N$; the dB cycle as given by Kaveh et al. eq. (20) at $d = 1$, Hindersin–Traulsen SI eq. (18) and Richter eq. (12), namely $2(r-1)/\big(3r - 1 + (r-3)r^{2-N}\big)$; the dB star (Allen eq. (2)).
- Hindersin & Traulsen, $N = 4$: their $C_4$ dB eqs. (13), (14) reproduced exactly; the four degree-heterogeneous graphs are Bd amplifiers and dB suppressors; $C_4$ is the only dB-"unclassified" graph; $K_4$ equals the well-mixed population. Stars $S_4..S_6$ are Bd amplifiers and dB suppressors on the $r$ grid.
- Richter: among the 265 quartic graphs on 11 vertices exactly one parent is an amplifier constructor, all four removals at its argmax vertex give $N_{\text{eff}} = 11.000756$ (printed 11.0008); among the 85 cubic graphs on 12 vertices exactly one constructor, two removals give 12.026644 (printed 12.0266; the third removal disconnects).

**Negative controls (all rejected as intended).**

- **Allen eq. (3) as printed in arXiv v1.** The cycle formula with $r^{-(N-1)}$ *disagrees with the exact chain in 25 of 25 cases*; three other sources and the exact chain agree with $r^{2-N}(r-3)$. arXiv v1 misprints the exponent; the published version (PLoS Comput. Biol. 2020, Eq. (3)) has $r^{-(N-3)}$ and agrees with the exact chain (the auditor confirmed this on $C_4, C_5, C_6$ at $r = 5/4, 3/4$, and that the published form agrees while the v1 form does not). So this is not an open discrepancy; the criterion used here (eqs. (5)–(9)) does not depend on eq. (3) (`AMC-C-0007`).
- A planted wrong random walk, $p_{ij} = 1/k_j$ in place of $1/k_i$, breaks the star value 10/3.
- $K_N$ never passes the strict test $N_{\text{eff}} > N$.
- The Antal approximation $N\mu_1^2/\mu_2$ is $\leq N$ on every connected graph with $N \leq 7$, so it can never detect an amplifier; it "fails" the Richter control by construction.
- Edges removed away from the argmax vertex: $N_{\text{eff}} \leq 10.8805 < 11$ on Richter's parent, $\leq 11.9337 < 12$ on the cubic parent.
- Lean (kernel decides the right way, `AMC-A-0004`): `ctl_perturbed_vector_rejected` (one $\tau$ entry changed by $1/D$ gives ¬Eq5), `ctl_perturbed_graph_rejected` (the R11 certificate is ¬Eq5 on the parent: a wrong transition matrix is caught), `ctl_parent_not_amplifier`.
- Lean (caught by the axiom audit): the planted `ctl_sorry` reports `sorryAx` and `ctl_axiom` reports `AmplifierCensusControls.ctl_cheat`, although `Controls.lean` compiles with rc 0 (a compile exit code alone accepts both).
- `ControlsMustFail.lean` asks the kernel to accept the perturbed vector and fails with rc 1 at elaboration ("did not reduce to isTrue or isFalse"), not by a kernel `false` verdict; it shows only that the claim is not provable this way. The decisive evidence is the kernel-checked, replayed `ctl_perturbed_vector_rejected`.
- A scan for `skipKernelTC`, `debug.`, `addDecl`, `#exit`, `implemented_by`, `native_decide` and `set_option` in the module is clean.

# 6 Lean formalisation

Modules in `formal_cert/Programmes/AmplifierCensus/` (copies in `data/lean/`): `Core.lean` (definitions `deg`, `p`, `Eq5`, `remeet`, `neff`; `eq5_unique`; `neff_of_cert`), `DataR11.lean` and `DataP11.lean` (integer certificate data proposed by the untrusted `gen_lean.py`), `Richter11.lean` (`R11_amplifier`, `P11_isothermal`, symmetry/degree checks), `Controls.lean`, `ControlsMustFail.lean` (nothing imports the last two). Imports: `Mathlib.Tactic` and `Mathlib.Algebra.BigOperators.Fin`; never `import Mathlib`; no `sorry` outside the planted control. Compiled through `scripts/certified_numerics/lean_pinned.py` (Lean 4.34.1, Mathlib `d13f23b7`). The in-file `#print axioms` reports, for `R11_amplifier`, `P11_isothermal` and `eq5_unique`, exactly [`propext`, `Classical.choice`, `Quot.sound`] (`data/logs/lean_richter11.json`); the producer's `leanchecker` replay named `Richter11` and `Controls` (rc 0; `data/logs/leanchecker_replay.json`), and the independent auditor's replay on oleans rebuilt from source in a private directory (byte-identical to the producer's) returned rc 0 on all five modules `Core`, `DataR11`, `DataP11`, `Richter11` and `Controls` (`data/leanchecker_replay_audit.json`; 57–74 s each). Replay does not check axioms (`Controls.lean` replays with rc 0 despite its planted cheats); the `#print axioms` output does. This used the pinned wrapper and not `anse/formal/lean_runner.py`, a deviation from the repository's usual route. sha256 of the files:

- `Core.lean`: `df9e30617ec88468cb18d986c2606665612bc528d19335fd10db07528f75b336`
- `DataR11.lean`: `050f91ac06bb7ba122c02c25e1b8295aa0c034b72588df740e24ac380ca66571`
- `DataP11.lean`: `cac552f026aec1cf36ebf91a010e41f6f03fc3bdf3747304b88f481bb5dff67b`
- `Richter11.lean`: `7d79537ab4e16f26c4a821f0c04a6935d97acbbf61c790bdc4a1cb7f740fa1ce`
- `Controls.lean`: `8f221b83258756f6d7ad6b7975f73da30d27d115797c0971f10dee9272fdf177`
- `ControlsMustFail.lean`: `d1230dd1bdae30c2840d165230807ef7f6d2b46d4c888310cc59acccde89bd3b`

**What Tier A certifies:** the stated linear-system theorems on the stated adjacency. **It does not certify** that eqs. (5), (6), (9) are Allen's equations as printed (a person must compare), that $N_{\text{eff}} > N$ implies amplification ([L]), the enumeration, or any census count.

# 7 Claim ledger

`ledger/build_ledger.py` files 22 claims: 4 at Tier A (`AMC-A-0001..0004`), 2 at L (`AMC-L-0001, 0002`), 14 at C (`AMC-C-0001..0014`, the last being the headline) and 2 at X (`AMC-X-0001`, the extrapolation; `AMC-X-0002`, recording that the repository guards were not run). Each has a content-addressed evidence blob in `ledger/evidence/`. In the data, some finite-$r$ statements are exact-rational certificates and are filed C as in the run note; the extrapolation beyond the certified points is X. The gate (`tools/ledger.py ledger.json --evidence-dir evidence`; verbatim in `ledger/gate_output.txt`) exits with code **1**: **0 blocking findings and 4 flags** (`LEDGER_UNAUDITED_TIER_A`, one for each Tier A row: no *person* has certified that the Lean statements mean what their glosses say). Exit code 1 is the honest state, not a failure to be cleared by editing. An independent AI audit (ACCEPT_WITH_CORRECTIONS, `AUDIT.md`) is recorded in an `ai_audit` field of each Tier A row; the gate field `audit` stays null because no person has audited the statements. There is no human audit and no statement lock.

# 8 Limitations

- **Novelty of the exhaustive statement is established only to our knowledge (a literature search, not a proof of absence).** We read Allen et al. (arXiv 1906.01036v1), Richter (2008.01446v1; 2204.12861v1, in pages), Tkadlec et al. (1906.02785v1), Hindersin & Traulsen (1504.03832v2), Kaveh et al. (1411.4556v1), and, for novelty only, Svoboda et al. (2401.14914v1) and Brewster et al. (2605.26411v3, Bd only), plus OEIS A001349 and A005177. Richter [3] (p. 8) reports only that his iterative *regular-input* procedure found none with $N \leq 10$; Svoboda et al. (p. 5) call Richter's graph "the smallest known". The independent auditor identified Richter's ref. [6] as Allen et al. [9] (PLoS Comput. Biol. 17:e1008695, 2021), whose exhaustive analysis of all connected graphs up to size 10 is for *Birth-death* updating only (read via the article page, not in full); the related Alcalde Cuesta et al. [10] database of all graphs of order $\leq 10$ and its companion "Suppressors of selection" (PLoS ONE 12:e0180549, 2017) are likewise Birth-death only (the chapter could not be opened). To our knowledge there is no prior exhaustive

$N \leq 10$ death-Birth weak-selection statement: the auditor's alphaXiv searches and its reading of ten papers (Allen 2019/2020 and 2021, Richter 2020 and 2022, Tkadlec 2019, Hindersin–Traulsen, Kaveh et al., Svoboda et al., Brewster et al. 2026 and arXiv:2512.11164) found none. Hindersin–Traulsen classified random graphs on an $r$ grid, which is not an exhaustive weak-selection census. This is a search result, not a proof of absence, and nothing here should be read as a priority claim.

- **The census is Tier C, not Lean.** Completeness depends on nauty `geng` (2.7r3, Debian, Apache-2.0) and on the match with OEIS A001349; the enclosure argument is a hand proof implemented in Python; the Lean theorems cover only the $N = 11$ linear system.
- **Scope of the statement.** Unweighted, undirected, simple, connected graphs; death-Birth updating; uniform initialisation (one mutant at a uniformly random vertex); the weak-selection criterion $N_{\text{eff}} > N$ at first order in $\delta$. Temperature or mutation-at-birth initialisation can change the classification (Hindersin & Traulsen, Richter 2022) and nothing transfers. Weighted transient amplifiers (Allen's Fans, Separated Hubs) are out of scope. A weak-selection amplifier is not an amplifier at finite $r$ and not a universal amplifier (Tkadlec et al., Thm. 1, as recorded in the run note, forbids that for non-complete graphs).
- **Finite-$r$ checks are not proofs for all $r$ near 1.** The four signs, the bisection and the 212 regular graphs are certified only at the listed $r$; uniqueness of the crossing is not shown; whether regular graphs are reducers for all $r$ near 1 remains open.
- **No human audit; no statement lock.** An independent AI audit agent (ACCEPT_WITH_CORRECTIONS; census rechecked by a different exact method for $N = 3..9$ in full and a 20 177-graph sample at $N = 10$, Lean rebuilt and replayed, finite-$r$ signs reproduced) has reviewed this paper, but no person has audited the Lean statements or the hand arguments, and no statement lock was registered for the new module. The $N = 10$ recheck is a sample, not a full independent census.
- **Repository guards were not run** (`antigravity_guard.py`, `test_rigor_guard.py`, `pytest tests/`); no guard output is claimed.
- **Deviations from the preregistration.** (1) The $N = 9, 10$ census is a certified enclosure plus exact solves of the candidates, not an exact rational solve of every graph. (2) The preregistration expected a Tier A certificate per listed amplifier; there is none at $N \leq 10$, so Tier A went to Richter's graph and its parent. (3) Finite-$r$ certificates use Python `Fraction`s, not python-flint (not installed), so they are Tier C, as preregistered for that route. (4) C1a (the $K_N$ and cycle closed forms) was checked as exact rational comparisons with the exact chain, not as Lean theorems. (5) C1x (birth/death-rate planes on tissue-like graphs) was **not attempted**. (6) The Lean gate used `lean_pinned.py`. (7) CPU accounting: at most two busy processes, but idle wrapper processes at times made three; some shell-guard lapses occurred (wait loops, one empty heredoc, one stray variable); none touched git.
- **Open discrepancies with Richter.** (i) Richter's printed $\tau_{\max}$ values (Section 3) are unexplained; the auditor ruled out the perturbed-graph hypothesis. (ii) Richter's $r_{\max} \approx 1.00075$ is inconsistent with our certified signs at 1.0007–1.0008125 (bracket $[1.00082333, 1.00082349]$); not investigated. Neither has been raised with the author. Allen eq. (3) is not open: arXiv v1 misprints it and the published version is correct.
- **Data note.** The run note quotes the cubic $\tau_{\max}$ as 18.29406; the data file gives 18.29410, used here.

# 9 Reproducibility

Repository AutoevolveAI, branch `worktree-openai-math-discovery`; HEAD when this paper was written: `25eacb7c37b7ebb54b53792cae13283a493e7bdf`; the C1 outputs are uncommitted working-tree files at that time (nothing was pushed or published). Copies of the inputs are in `data/` (`data/code/` the scripts, `data/lean/` the Lean sources, `data/logs/` compile and census logs, `data/RESULT.md` and `data/result.json` the run note). The graph6 files of the census ($10^7$ graphs at $N = 10$) are not included; their sha256 values are in `data/result.json`. Commands (tested in this revision on $N = 7$ and on the Richter graph; paths relative to the repository root, then to a work copy `W`):

```
# Work directory W: the scripts write next to their own folder, so run them from a copy, not in place.
mkdir -p W/code W/data ; cp papers/p4_amplifier_census/data/code/*.py W/code/ ; cp papers/p4_amplifier_census/data/*.g6 W/ ;
     cd W
nauty-geng -cq 10 g10.g6                                        # quick test: N = 7 (853 graphs, g7.g6, enc7.json)
python -I code/census_enclosure.py g10.g6 10 0 1 enc10.json    # <graph6 file> <N> <shard> <nshards> <out.json>
python -I code/finish_candidates.py enc10.json census_n10.json
python -I code/exact_census.py g7.g6 7 census_n7_exact.json enc7.json  # (N <= 8 only) independent Fraction census
python -I code/controls.py ; python -I code/richter.py         # controls (data/controls.json), Richter reproduction
python -I code/fixation_cert.py richter_n11_amplifier.g6 fc.json 10001/10000 2499/2500
python -I code/window_bisect.py richter_n11_amplifier.g6 wb.json 10007/10000 20017/20000 10
python -I code/regular_check.py reg.json
python -I code/gen_lean.py richter_n11_amplifier.g6 R11 DataR11.lean   # proposes the Lean data module
# from the repository root:
python -I scripts/certified_numerics/lean_pinned.py compile formal_cert/Programmes/AmplifierCensus/Richter11.lean
python -I papers/p4_amplifier_census/data/worked_example.py out.json   # the exact examples of Step 3
python -I papers/p4_amplifier_census/ledger/build_ledger.py ; python -I papers/p4_amplifier_census/ledger/run_gate.py
```

Not re-run: the full $N = 8..10$ census and `run_small.py`/`summarize.py`/`leanchecker_replay.py`, which assume the original C1 directory layout (see `REVISION.md`). The independent auditor's fresh `nauty-geng -cq 10` output has sha256 `bf530d38...0ab5`, equal to the hash the producer recorded (`data/result.json`). The run used cores 4 and 5 (taskset) and no GPU; the $N = 10$ enclosure took 1237 s on one core.

# References


[1] B. Allen, C. Sample, R. Jencks, J. Withers, P. Steinhagen, L. Brizuela, J. Kolodny, D. Parke, G. Lippner, Y. A. Dementieva, *Transient amplifiers of selection and reducers of fixation for death-Birth updating on graphs*, PLoS Comput. Biol. 16(1):e1007529 (2020), arXiv:1906.01036 (v1 read; eqs. (5)–(9) and the definitions are identical in the published version, pp. 4–6, checked by the auditor). Source of the criterion $N_{\mathrm{eff}} > N$ ([L]).

[2] H. Richter, *Constructing transient amplifiers for death-Birth updating: A case study of cubic and quartic regular graphs*, arXiv:2008.01446 (v1, 4 Aug 2020); the eleven-vertex graph (Fig. 1a) and the cubic twelve-vertex example. The journal version appears to be *Spectral analysis of transient amplifiers for death–birth updating constructed from regular graphs*, J. Math. Biol. 82, 61 (2021) (as cited by Svoboda et al. and Richter 2022).

[3] H. Richter, *Spectral dynamics of guided edge removals and identifying transient amplifiers for death-Birth updating*, arXiv:2204.12861 (2022); cited by Svoboda et al. as J. Math. Biol. 87(1):3 (2023).

[4] J. Tkadlec, A. Pavlogiannis, K. Chatterjee, M. A. Nowak, *Limits on amplifiers of natural selection under death-Birth updating*, PLoS Comput. Biol. 16(1):e1007494 (2020), arXiv:1906.02785.

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[8] D. A. Brewster, G. Lippner, J. Tkadlec, M. A. Nowak, *Fixation location in structured populations*, arXiv:2605.26411 (v3, 28 Jun 2026); Birth-death only, read for novelty only.

[9] B. Allen, C. Sample, P. Steinhagen, J. Shapiro, M. King, T. Hedspeth, M. Goncalves, *Fixation probabilities in graph-structured populations under weak selection*, PLoS Comput. Biol. 17(2):e1008695 (2021); exhaustive analysis for Birth-death updating (read via the article page).

[10] J. Alcalde Cuesta, P. González Sequeiros, Á. Lozano Rojo, *An accurate database of the fixation probabilities for all undirected graphs of order 10 or less*, IWBBIO 2017, LNCS 10209, pp. 209–220; cited by Richter 2022 as ref. [2]; not opened.

[11] X. Callens, *The isothermal theorem of evolutionary graph theory for birth-death updating, kernel-checked in Lean 4: Moran fixation on strongly connected isothermal graphs and its stationary-selection generalization*, Zenodo, doi:10.5281/zenodo.23660071 (2026). Own preprint, the predecessor of this one; not peer reviewed.

[12] B. D. McKay, A. Piperno, *Practical graph isomorphism, II*, J. Symb. Comput. 60 (2014) 94–112; nauty/`geng` 2.7r3 used.

[13] OEIS Foundation, sequences A001349 (connected graphs on $n$ nodes) and A005177 (connected regular graphs), `https://oeis.org`, read 2026-10-27.

[14] The mathlib Community, *The Lean mathematical library*, arXiv:1910.09336 (2019); cited for the library only.